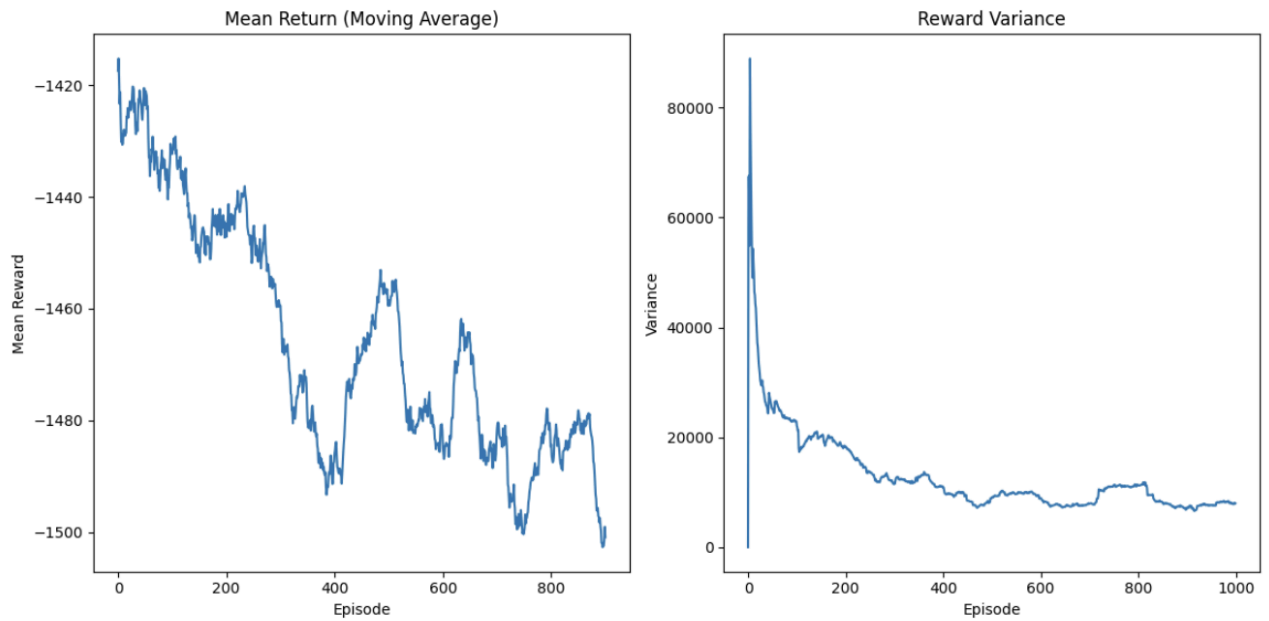
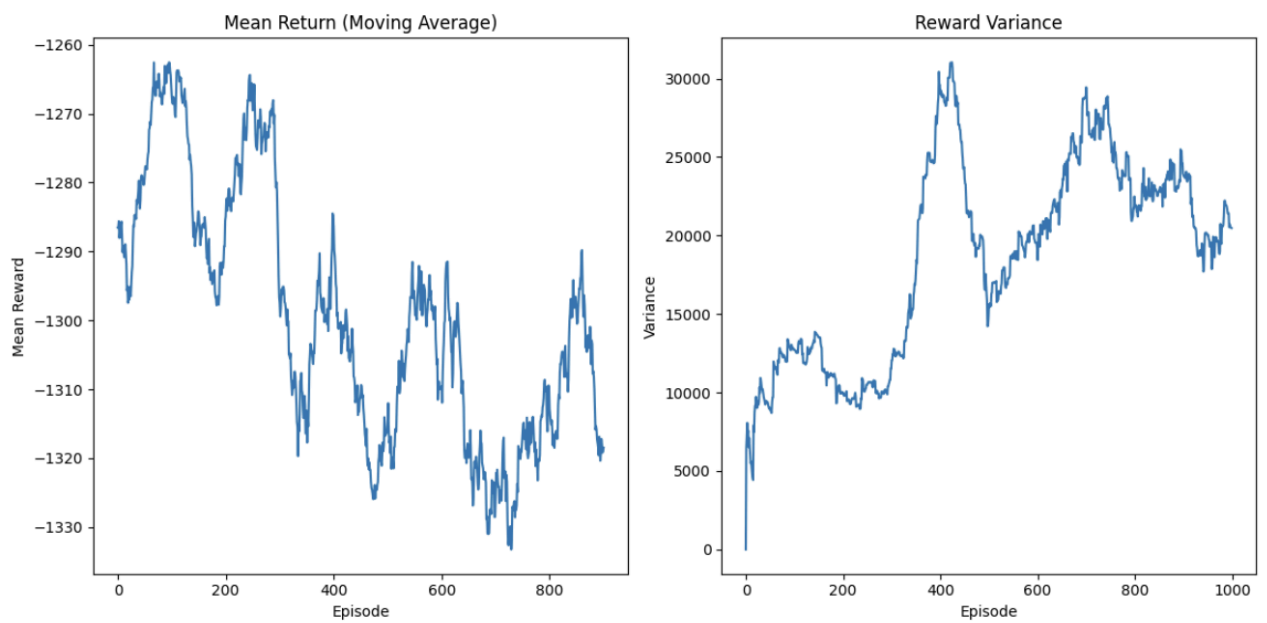


VANILLA DQN ALGORITHM



DQN ALGORITHM WITH NOISY EXPLORATION



Mean Return (Moving Average) Comparison:

- **Vanilla DQN** : The mean reward tends to decrease over episodes, indicating that the agent struggles to improve learning effectively. It also exhibits fluctuations, suggesting instability in learning.
- **Noisy Exploration DQN** : The mean reward is higher than in Vanilla DQN, indicating better learning performance. The fluctuations are still present but less steep than in Vanilla DQN.

Reward Variance Comparison:

- **Vanilla DQN**: Initially, the variance is extremely high, but it decreases significantly over time. This suggests that the agent initially explores a wide range of actions but later converges to a more stable policy.
- **Noisy Exploration DQN**: The variance remains higher for a longer period, indicating that noisy exploration continues throughout training. This can lead to better exploration and potentially better final performance, though it introduces instability.

Observations:

- Noisy Exploration DQN achieves a higher mean reward, which implies **better overall learning**.
- The variance in Noisy Exploration DQN remains high, meaning it **explores more** than Vanilla DQN, which can help avoid local optima.
- Vanilla DQN shows a **steady decline** in performance, whereas Noisy DQN maintains a **better performance trend**.

Conclusion:

Noisy Exploration DQN appears to **outperform** Vanilla DQN by maintaining higher rewards and better exploration. However, it introduces **more variance**, which could make training less stable. If tuned properly, Noisy Exploration DQN can help the agent learn more effectively.